

CHAPTER 3: RESULTS

3.1 Demographic and Occupational Profile

A total of 376 wood industry workers participated in the study, representing an 87.5% response rate from 430 initially approached. The mean age of participants was 34.69 ± 11.46 years (range: 18–60 years), and the mean duration of employment in the wood industry was 11.74 ± 9.67 years, indicating a sample that ranged from relatively recent entrants to seasoned professionals with substantial cumulative occupational exposure.

Table 3.1: Socio-demographic and Occupational Characteristics of Study Participants

Variable	Category	n	Percentage %
Smoking status	Yes	91	24.2
	No	285	75.8
Education	Illiterate	95	25.3
	Primary	113	30.1
	Secondary	146	38.8
	Higher	19	5.1
Marital Status	Single	104	27.7
	Married	272	72.3
PPE usage	Never	124	33.0
	Sometimes	161	42.8
	Always	91	24.2
Workplace ventilation	Present	315	83.8
	Absent	59	15.7

The majority of participants had attained modest levels of formal education: 38.8% had completed secondary schooling, 30.1% primary education, 25.3% were entirely illiterate, and a mere 5.1% had achieved higher education. This educational profile carries direct occupational safety implications, as limited literacy may impair workers' capacity to interpret hazard warning labels, follow written safety protocols, and engage effectively with formal training programs. Approximately one in four workers (24.2%) were current smokers — a proportion of particular significance given the synergistic respiratory risks conferred by simultaneous tobacco and wood dust exposure.

Regarding institutional safety infrastructure, 58.8% of workers reported having received formal occupational safety training, and 58.0% had been provided with PPE. However, actual PPE utilization was markedly inconsistent: only 24.2% reported always using protective equipment, 42.8% used it sometimes, and 33.0% never used it. This pronounced compliance gap between provision and consistent utilization represents a critical programmatic finding. It highlights that physical availability alone cannot overcome behavioral or ergonomic barriers to safety compliance, such as equipment discomfort or a workplace culture that normalizes risk-taking. The majority of workplaces (83.8%) were reported to have some form of ventilation system, while 15.7% of workers operated in poorly ventilated environments. For these individuals, the absence of primary engineering controls maximizes continuous breathing-zone exposure to suspended particulates, directly elevating their localized occupational risk.

the demographic profile of this cohort highlights a profound socioeconomic vulnerability that compounds their occupational health risks. The intersection of limited formal education, informal employment arrangements, and hazardous working conditions creates a multi-dimensional burden for these laborers. Notably, the financial precarity of this demographic is starkly reflected in their healthcare financing; a majority of the workforce (52.4%) bears the cost of medical treatment entirely out-of-pocket, with only a negligible fraction (8.2%) possessing formal health insurance. In an environment where over half the workforce has required time off due to occupational trauma, this lack of financial protection creates a vicious cycle where workplace morbidity directly precipitates economic hardship. Laborer continue to operate heavy machinery and endure particulate exposure despite actively suffering from chronic musculoskeletal pain or respiratory distress, simply out of financial necessity. Ultimately, this demographic snapshot reveals a

workforce that is not only highly exposed to industrial hazards, but also systematically underequipped both educationally and economically to advocate for safer conditions or access restorative healthcare.

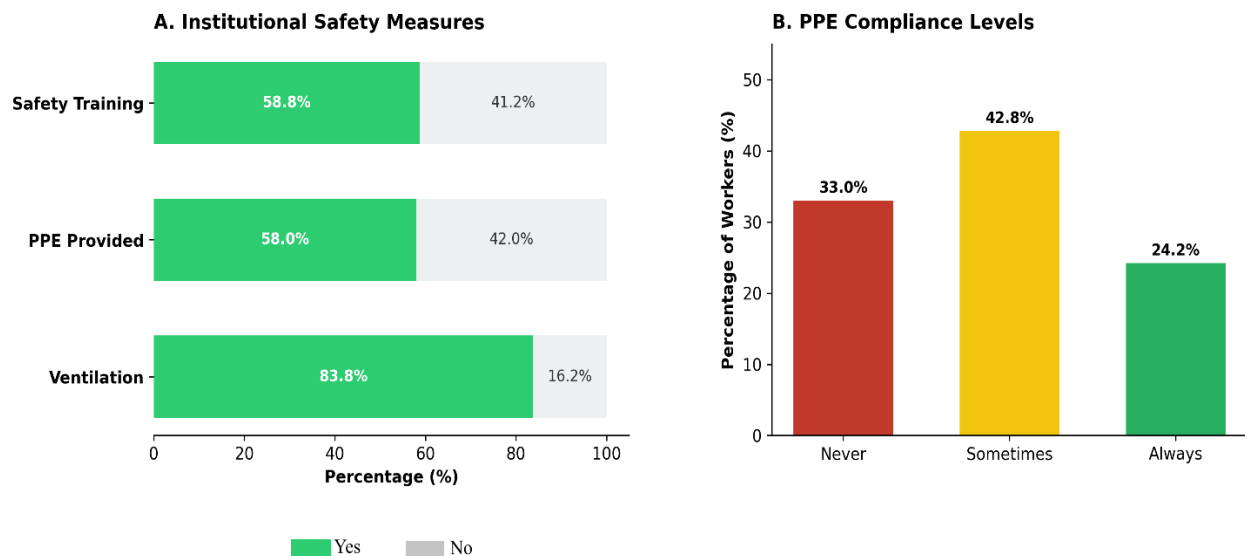


Figure 2: Institutional Safety and PPE Compliance

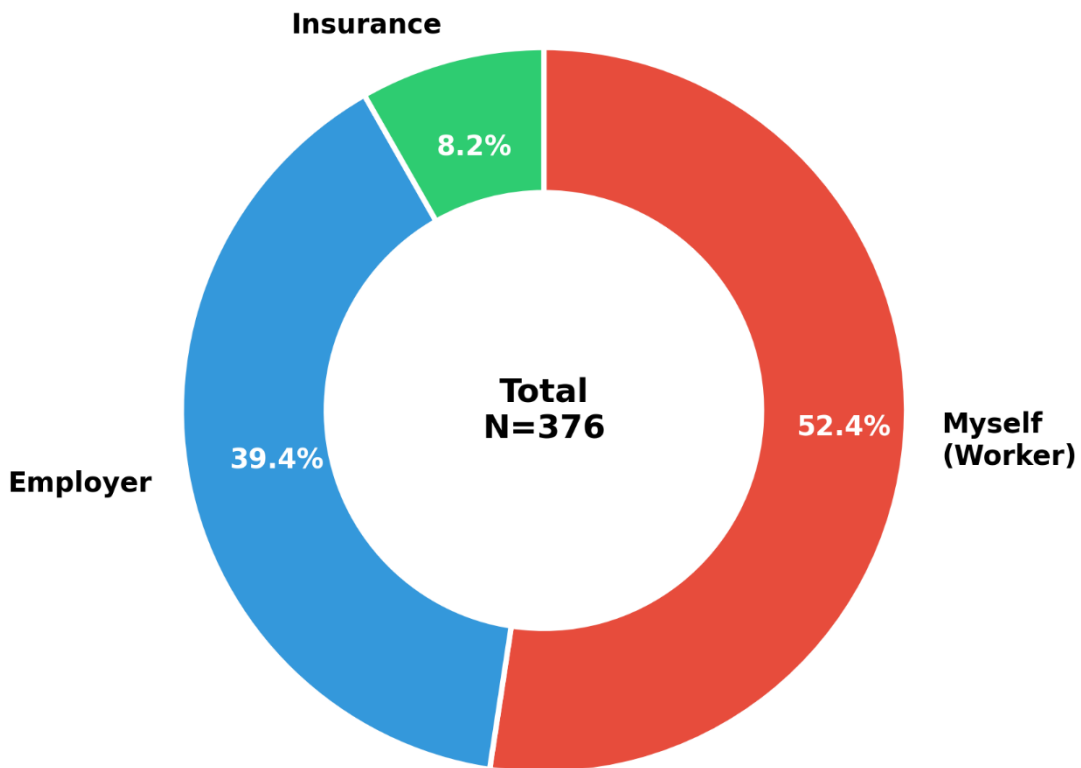


Figure 3: Medical Treatment Coverage

Notably, 52.4% of workers bear medical expenses personally, 39.4% are covered by employers, and only 8.2% have insurance coverage, reflecting the precarious healthcare financing situation of this workforce.

3.2 Hazard Perception, Safety Culture, and Injury Prevalence

Table 3.2: Hazard Perception, Safety Culture, and Injury Prevalence

Variable	Category	Frequency (n)	Percentage (%)
Hazard Awareness:			
Dust	Yes	288	76.6
	No	37	9.8
	I don't know	51	13.6
Hazard Awareness:			
Noise	Yes	233	62
	No	93	24.7
	I don't know	50	13.3
Machine Safety Guards	Yes, machinery has adequate guards	235	62.5
	No, machinery lacks safety guards	115	30.6
	Not applicable to my job	26	6.9
Ergonomic Autonomy (Breaks)	Yes, can take breaks without problem	141	37.5
	Yes, but must inform supervisor	175	46.5
	No, breaks are only at fixed times	50	13.3
	No, there are very few breaks	10	2.7
Confidence Reporting			
Safety	Very confident	116	30.9

	Somewhat confident	170	45.2
	Not at all confident	90	23.9
Time Off Due to			
Work Injury	Yes	216	57.4

To better understand the behavioral and organizational context of the Hayatabad Industrial Estate, workers were assessed on their perception of occupational hazards, workplace safety culture, and recent injury history (Table 3.2).

A considerable proportion of the workforce demonstrated a high awareness of specific physical hazards. The majority of workers (76.6%) correctly identified wood dust as a significant health risk. However, awareness regarding auditory hazards was notably lower, with only 62.0% recognizing the harmful effects of occupational noise. This leaves over a third of the workforce either unaware or unsure of noise-induced risks.

Despite general hazard awareness, the institutional safety culture revealed significant vulnerabilities. When asked about their confidence in reporting safety problems or hazards to management, only 30.9% of workers felt "Very confident." The largest segment (45.2%) felt only "Somewhat confident," and nearly one-quarter of the workforce (23.9%) reported feeling "Not at all confident" in bringing safety issues to light.

This precarious safety culture is reflected in a high rate of occupational trauma; more than half of the surveyed workforce (57.4%) reported having to take time off work due to a workplace injury.

Assessment of the physical and ergonomic workplace environment revealed critical gaps in primary safety infrastructure. Nearly one-third of the workforce (30.6%) reported operating machinery that lacked adequate safety guards or protective features. Regarding ergonomic autonomy and physical recovery, only 37.5% of workers felt they could take short breaks freely without issue. The largest proportion of the workforce (46.5%) was required to obtain explicit supervisory permission to rest, and 16.0% operated under rigid or highly restricted break

schedules. This lack of ergonomic autonomy substantially compounds the cumulative musculoskeletal loading observed in this cohort

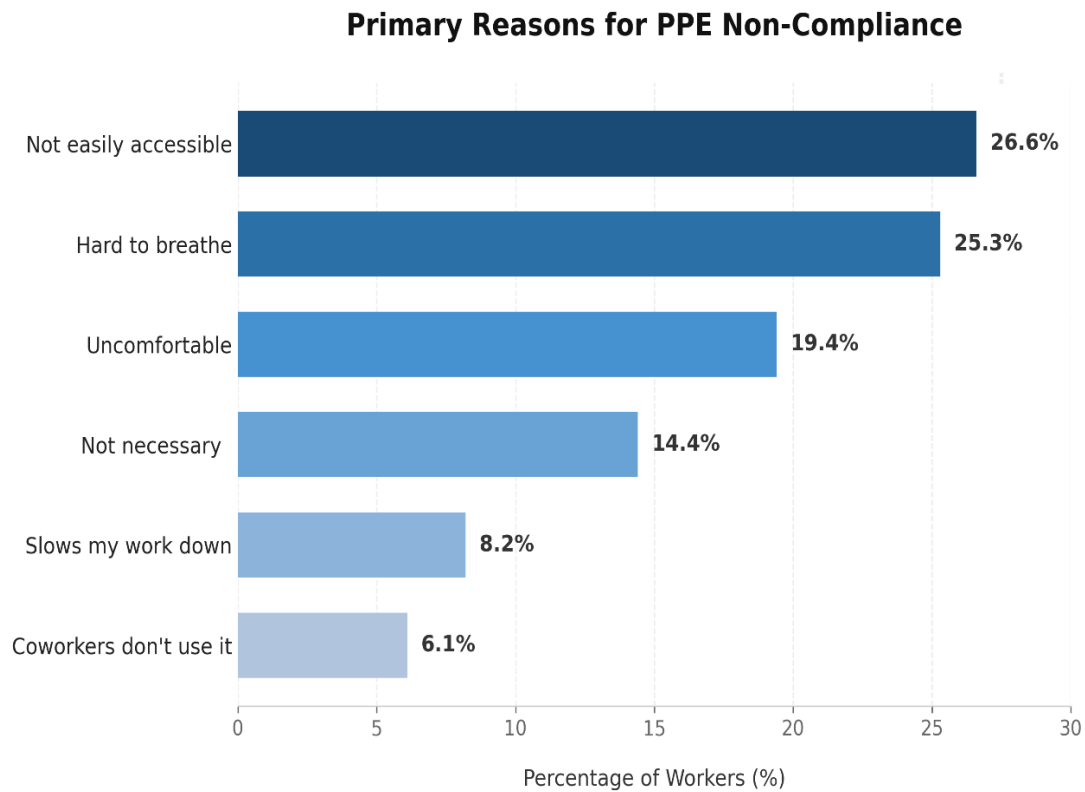


Figure 4: Primary Barriers to PPE Utilization Among Woodworkers

This bar chart illustrates the distribution of self-reported primary obstacles to the consistent use of Personal Protective Equipment (PPE) among the study cohort at the Hayatabad Industrial Estate.

3.3 Prevalence of Occupational Health Symptoms

Health symptoms were dichotomized into Low Symptom Burden (Never or Rarely reported) and High Symptom Burden (Sometimes or Often reported) to facilitate clinical interpretation and statistical analysis. The distribution of symptom frequencies across all six health domains measured in this study is presented in Figure 4.

Musculoskeletal pain emerged as the most prevalent occupational health problem, with 38.0% of workers (n = 143) reporting high-frequency symptoms. Respiratory cough was the second most prevalent condition (34.8%, n = 131), followed by headaches (27.1%, n = 102), eye irritation (23.9%, n = 90), skin problems (21.0%, n = 79), and tinnitus (17.6%, n = 66). The simultaneous co-occurrence of multiple symptoms within individual workers is consistent with the multi-hazard and cumulative exposure environment characteristic of wood processing settings, and underscores the importance of adopting comprehensive — rather than single-hazard — occupational health surveillance strategies.

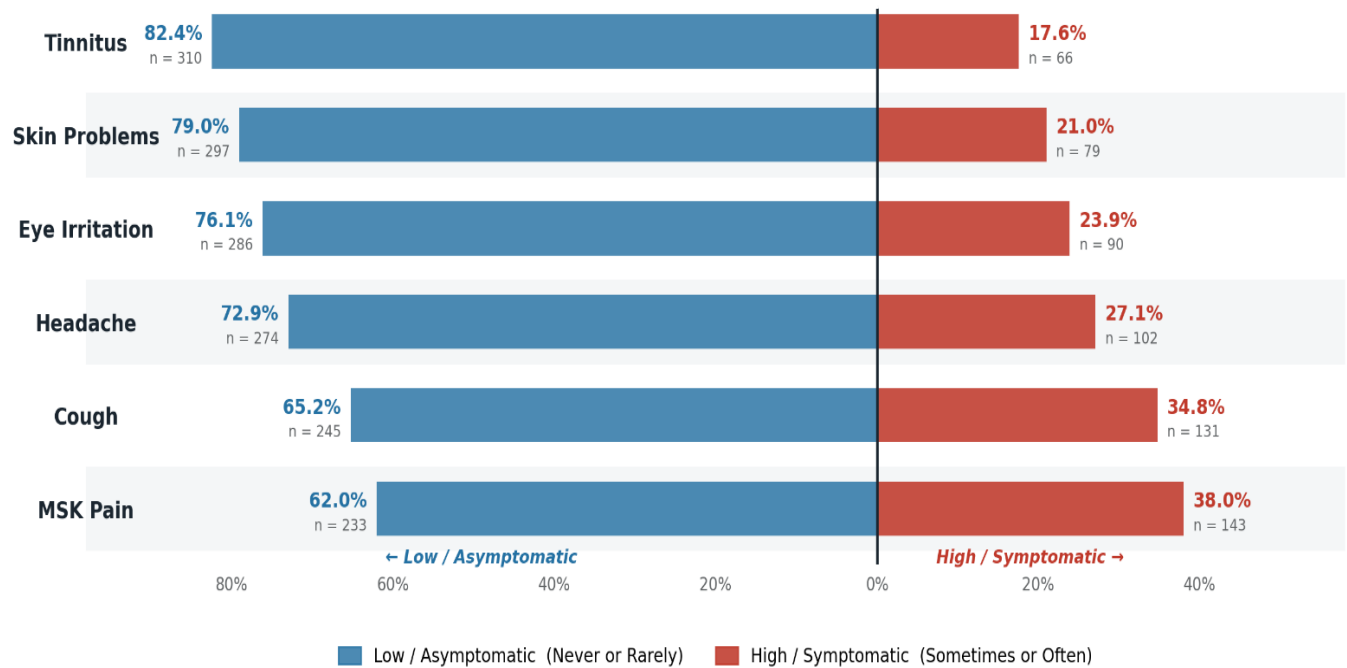


Figure 5: Prevalence of occupational health symptoms

3.4 Association Between PPE Usage and Health Symptoms

Chi-square analyses were performed to assess the statistical association between PPE usage frequency and each of the six health outcomes.

Table 3.4: Chi-Square Analyses — Association Between PPE Usage Frequency and Health Outcomes

Health Outcome	PPE: Never	PPE: Sometimes	PPE: Always	Total	Symptom Frequency	χ^2	df	p
Respiratory Cough	57	128	60	245	Low	34.733	2	< 0.001
	67	33	31	131	High			
Eye Irritation	84	126	76	286	Low	7.920	2	0.019
	40	35	15	90	High			
Skin Problems	85	134	78	297	Low	12.369	2	0.002
	39	27	13	79	High			
Musculo-skeletal Pain	66	102	65	233	Low	7.608	2	0.022
	58	59	26	143	High			
Tinnitus	88	139	83	310	Low	17.800	2	< 0.001
	36	22	8	66	High			
Headache	87	122	65	274	Low	1.244	2	0.537
	37	39	26	102	High			

PPE usage demonstrated statistically significant associations with five of the six health outcomes examined in this study, with the sixth — headache — falling short of statistical significance ($\chi^2 =$

1.244, $p = 0.537$). Among the significant associations, the strongest effect was observed for respiratory cough ($\chi^2 = 34.733$, $p < 0.001$), followed by tinnitus ($\chi^2 = 17.800$, $p < 0.001$), skin problems ($\chi^2 = 12.369$, $p = 0.002$), eye irritation ($\chi^2 = 7.920$, $p = 0.019$), and musculoskeletal pain ($\chi^2 = 7.608$, $p = 0.022$). Across all five significant outcomes, the distribution of symptom severity shifted notably toward lower frequency among workers reporting consistent PPE use, which lends face validity to the protective role of personal protective equipment in this occupational setting.

It is important, however, to interpret these bivariate associations with appropriate methodological caution. Chi-square analyses, by their nature, establish co-occurrence rather than causation, and the cross-sectional design of this study precludes any inference about temporality or directionality. A particular concern here is the possibility of reverse causality: workers assigned to higher-risk job roles — and thereby experiencing a greater symptom burden — may simultaneously be issued and observed wearing PPE more frequently as a direct consequence of institutional risk stratification or supervisory oversight. In other words, the relationship between PPE use and symptom severity may, in part, run in the opposite direction to what a straightforward protective interpretation would suggest. Additionally, unmeasured confounders such as duration of occupational exposure, workshop ventilation quality, and pre-existing health conditions could plausibly explain some portion of the observed variation. These limitations inherent to bivariate analysis underscore the necessity of the multivariate regression models presented in Section 3.6, which allow for simultaneous adjustment across relevant covariates and offer a more robust basis for causal inference.

3.5 Association Between Ventilation Status and Health Symptoms

Table 3.5: Chi-Square Analysis — Association Between Workplace Ventilation and Respiratory/Eye Symptoms

Outcome	Ventilation	Never	Rarely	Sometimes	Often	χ^2 (p-value)
Cough	Present (n=315)	128	90	69	28	22.92 (0.001)**
	Absent (n=59)	7	18	25	9	
Eye Irritation	Present (n=315)	154	85	44	32	11.27 (0.080) ns

Absent (n=59)	23	23	12	1
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Workers in poorly ventilated workshops reported significantly higher frequencies of respiratory cough symptoms ($\chi^2 = 22.92$, $p = 0.001$), while the association between ventilation status and eye irritation did not achieve statistical significance ($\chi^2 = 11.27$, $p = 0.080$). The significant association with respiratory symptoms is consistent with the established pathophysiological role of inadequate ventilation in increasing airborne particulate and chemical concentrations within the breathing zone, thereby amplifying mucosal irritation and bronchoconstriction. The non-significant finding for eye irritation may reflect the multifactorial etiology of ocular symptoms in this setting, which may include direct mechanical particle contact not amenable to ventilation modification alone.

3.6 Association Between Work Duration and Health Symptoms

Table 3.6: Spearman's Rank Correlations — Cumulative Work Duration and Health Outcomes

Health Outcome	rs	p-value	Interpretation
Musculoskeletal Pain	0.308	< 0.001	Moderate positive correlation — significant
Headache	0.107	0.039	Weak positive correlation — significant
Eye Symptoms	0.085	0.101	Not significant
Tinnitus	0.067	0.197	Not significant
Skin Problems	-0.073	0.156	Not significant
Cough	0.038	0.460	Not significant

Cumulative work duration demonstrated a statistically significant moderate positive correlation with musculoskeletal pain ($r_s = 0.308$, $p < 0.001$), indicating that workers with longer employment tenures face progressively greater MSK symptom burden — a pattern consistent with the biological plausibility of cumulative musculoskeletal loading over time. A weaker but statistically significant positive correlation was also observed with headache frequency ($r_s = 0.107$, $p = 0.039$), potentially reflecting the cumulative neurological and vascular effects of chronic occupational noise and physical stress. Notably, no significant correlations with work duration were identified for cough, eye irritation, tinnitus, or skin problems, suggesting that these outcomes are

driven more acutely by exposure intensity at any given time point than by cumulative tenure.

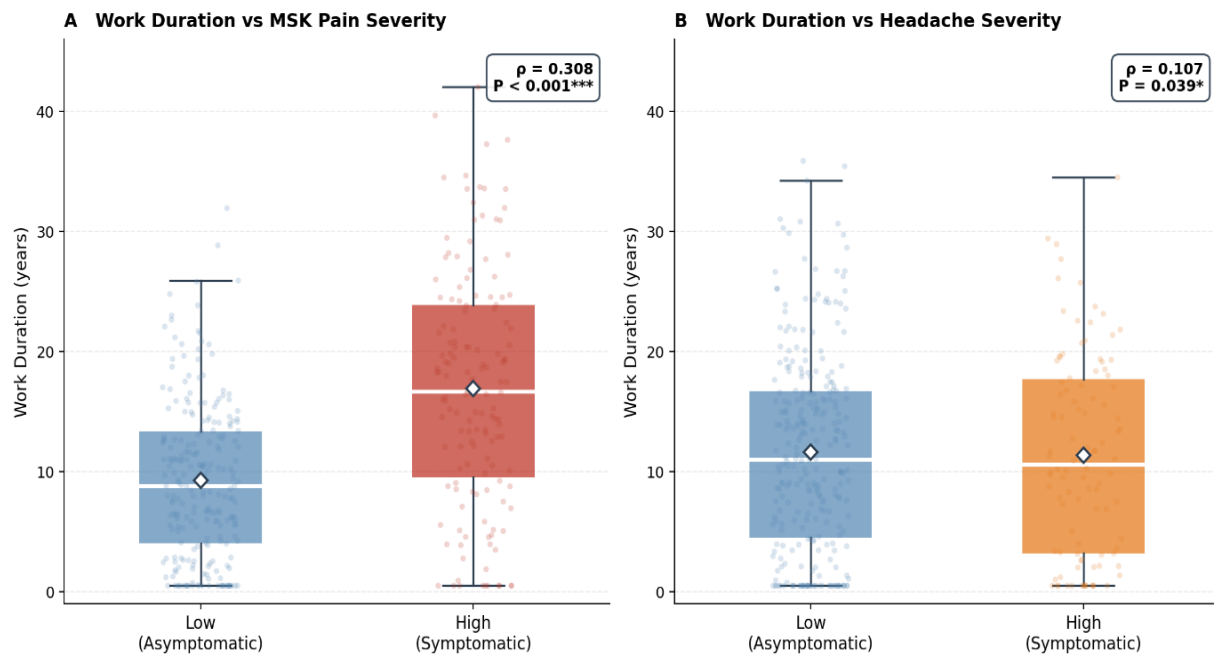


Figure 6: Box plots illustrating the association between cumulative work duration and symptom severity

3.7 Association Between Job Role and Health Symptoms

Chi-square analyses were performed to assess the association between job role and each of the six health outcomes. Results are presented in Tables 3.6.1 through 3.6.3.

Table 3.7.1: Chi-Square Analysis — Association Between Job Role and Health Outcomes

Health Outcome	χ^2	df	p-value	Significance
Respiratory Cough	22.687	8	0.004	Significant
Musculoskeletal Pain	21.004	8	0.007	Significant
Skin Symptoms	15.180	8	0.056	Borderline
Eye Symptoms	4.638	8	0.795	Not significant
Tinnitus	5.067	8	0.750	Not significant
Headache	7.324	8	0.502	Not significant

Job role was significantly associated with respiratory cough ($\chi^2 = 22.687$, $df = 8$, $p = 0.004$) and musculoskeletal pain ($\chi^2 = 21.004$, $df = 8$, $p = 0.007$), reflecting the differential exposure profiles inherent to different woodworking occupations. Skin symptoms approached but did not reach conventional significance ($p = 0.056$), while eye symptoms, tinnitus, and headache showed no statistically significant variation by job role.

Table 3.7.2: Job Role by Respiratory Cough Symptom Burden

Job Role	Low Symptom Burden	High Symptom Burden	Total
Machine Operator	47	43	90
Carpenter	38	28	66
Saw Operator	52	21	73
Helper	43	14	57
Supervisor	14	12	26

Spray Painter	20	4	24
Packing	15	3	18
Loader	7	5	12
Wood Polisher	9	1	10
Total	245	131	376

Machine operators exhibited the highest absolute count of high cough burden (n = 43 of 90; 47.8%), followed by carpenters (n = 28 of 66; 42.4%) and supervisors (n = 12 of 26; 46.2%). Saw operators and spray painters showed comparatively lower proportional burdens despite working in dusty and chemically exposed environments, which may reflect confounding by age, PPE practices, or specific task patterns within those roles. The overall significant chi-square result (p = 0.004) confirms that respiratory symptom prevalence is not uniformly distributed across job categories but varies meaningfully with occupational function.

Table 3.7.3: Job Role by Musculoskeletal Pain Symptom Burden

Job Role	Low Symptom Burden	High Symptom Burden	Total
Machine Operator	62	28	90
Carpenter	38	28	66
Saw Operator	46	27	73
Helper	39	18	57
Supervisor	8	18	26
Spray Painter	13	11	24
Packing	13	5	18
Loader	5	7	12
Wood Polisher	9	1	10
Total	233	143	376

The distribution of musculoskeletal symptom burden across job roles reveals a striking finding: supervisors demonstrated the highest proportional prevalence of high MSK burden (18 of 26; 69.2%), substantially exceeding the study-wide average of 38.0%. This finding is counterintuitive if MSK disorders are assumed to be solely a function of acute physical exertion, and may instead reflect the cumulative effect of prolonged static postures, sustained workstation-based tasks, or

extended working tenure characteristic of supervisory roles. Loaders also exhibited a notably high proportional MSK burden (7 of 12; 58.3%), consistent with the biomechanical demands of repetitive heavy manual lifting. Carpenters (42.4%) and spray painters (45.8%) showed above-average proportional burdens, while wood polishers had the lowest (1 of 10; 10.0%), possibly reflecting the relatively small sample size of that subgroup.

3.8 Independent-Samples T-Test: Continuous Predictors of Health Outcomes

Independent-samples t-tests were conducted to identify continuous variables demonstrating significant mean differences between workers with low and high symptom burden, prior to entry into binary logistic regression models. The complete t-test results for all six health outcomes and all continuous predictors are presented in the Supplementary Material (Table S1). Selected significant findings are summarized in Table 3.7 below.

Table 3.8: Summary of Significant Independent-Samples T-Test Results

Outcome	Predictor	Mean (Low)	Mean (High)	t	p-value
<i>Cough</i>	Smoking status	0.31	0.46	2.618	0.009
	Age (years)	33.58	36.77	−2.592	0.010
	Dust exposure score	5.42	5.06	2.461	0.014
	Standing hours/day	5.96	5.11	3.414	0.001
	PPE Usage	2.01	1.73	3.582	0.001
<i>Skin Problems</i>	Work duration (years)	12.26	9.80	2.017	0.044
	Dust exposure score	5.42	4.82	3.553	< 0.001
	Noise exposure score	6.09	5.46	2.265	0.024
<i>MSK Pain</i>	Age (years)	31.50	39.90	−7.369	< 0.001
	Work duration (years)	9.19	15.89	−6.913	< 0.001
	Standing hours/day	5.86	5.33	2.170	0.031

	PPE usage score	0.41	0.29	2.143	0.033
<i>Tinnitus</i>	PPE usage score	0.20	0.09	2.539	0.012
<i>Headache</i>	Age (years)	33.90	36.82	-2.212	0.028
	Noise exposure score	6.14	5.47	2.606	0.010

The t-test results reveal several clinically and statistically compelling patterns. For respiratory cough, smokers demonstrated a markedly higher smoking proportion in the high-burden group (mean score 0.46 vs. 0.31; $p = 0.009$), while workers with high cough burden were significantly older (mean 36.77 vs. 33.58 years; $p = 0.010$). An unexpected finding was that higher dust exposure scores and more standing hours were paradoxically associated with lower cough burden — a pattern potentially attributable to survivorship bias, whereby workers in the most heavily dust-exposed or physically demanding roles who remain in the workforce may represent a selected, physiologically resilient sub-population.

For musculoskeletal pain, the magnitude of the mean difference in age (31.50 vs. 39.90 years; $t = -7.369$; $p < 0.001$) and work duration (9.19 vs. 15.89 years; $t = -6.913$; $p < 0.001$) between low- and high-burden groups represents the largest and most clinically significant contrasts observed across all t-test analyses in this study. These effect sizes — exceeding 8 years in age and 6.7 years in work tenure — provide robust support for the biological plausibility of cumulative musculoskeletal loading as the dominant driver of MSK symptom burden in this population.

For skin problems, workers with higher perceived dust exposure scores showed significantly greater skin symptom burden (mean 5.42 vs. 4.82; $p < 0.001$), consistent with established dermatological sensitization mechanisms from chronic wood dust contact. Tinnitus was significantly associated with lower PPE usage scores in symptomatic workers (0.09 vs. 0.20; $p = 0.012$), underscoring the direct protective function of hearing protection devices. For headache, both older age and higher noise exposure scores were significantly associated with greater symptom burden, supporting a cumulative neurological stress model.

3.9 Binary Logistic Regression Analyses

Initially, ordinal logistic regression was planned to assess the full 4-point gradient of symptom severity (Never, Rarely, Sometimes, Often). However, substantial cell sparsity was observed (74.2% of cells with zero frequencies), and the proportional odds assumption was violated ($p = 0.047$), compromising model stability. To resolve these mathematical limitations, the outcome variable was dichotomized into a 2-point binary scale (Low vs. High symptom burden). Binary logistic regression was therefore adopted as the final multivariate model, as it successfully eliminates cell sparsity concerns, satisfies all core statistical assumptions, and produces highly stable, clinically interpretable estimates (Odds Ratios)

Table 3.9.1: Binary Logistic Regression — Predictors of Respiratory Cough

Predictor	Crude OR (95%CI)	p-value	AOR (95% CI)	p-Value
Smoking	1.89 (1.17 – 3.06)	0.010*	1.41 (0.84 – 2.38)	0.198
Age (years)	1.03 (1.01 – 1.04)	0.011*	1.01 (0.99 – 1.03)	0.327
Dust exposure				
score	0.82 (0.70 – 0.96)	0.015*	0.90 (0.75 – 1.06)	0.209
PPE Usage	0.59 (0.44 – 0.79)	<0.001*	0.64 (0.47 – 0.87)	0.004*
Standing				
Hours	0.85 (0.77 – 0.94)	0.001*	0.90 (0.81 – 1.00)	0.043

A comprehensive binary logistic regression analysis was conducted to identify the independent predictors of high respiratory cough burden among the study population. The overall multivariate model demonstrated a robust and statistically significant improvement in fit over the null model ($\chi^2 = 27.256$, $df = 5$, $p < 0.001$), explaining approximately 9.6% of the variance in cough symptom severity (Nagelkerke $R^2 = 0.096$). In contrast to preliminary models, the fully adjusted analysis revealed that occupational and behavioral factors superseded lifestyle factors as the primary drivers of respiratory health. Personal Protective Equipment (PPE) usage emerged as the strongest independent protective factor; for every one-unit increase in the frequency of PPE utilization, the odds of reporting a high cough burden decreased by 36% (AOR = 0.64; 95% CI: 0.47–0.87; $p = 0.004$). Additionally, prolonged standing hours demonstrated a significant inverse

relationship with symptom severity, where each additional hour of standing reduced the odds of high cough burden by 10% (AOR = 0.90; 95% CI: 0.81–1.00; $p = 0.043$). Notably, while smoking, age, and total dust exposure exhibited significant bivariate associations with cough severity in crude analyses, none achieved statistical significance after multivariate adjustment ($p > 0.05$). The adjusted odds ratio for smoking was reduced to a non-significant trend (AOR = 1.41; 95% CI: 0.84–2.38; $p = 0.198$), indicating that the raw effect of smoking is heavily confounded by the variance in protective workplace behaviors.

3.9.2 Predictors of Musculoskeletal Pain

Table 3.9.2: Binary Logistic Regression — Predictors of Musculoskeletal Pain

Predictor	Crude OR (95% CI)	p-value	AOR (95% CI)	p-Value
Age (years)	1.07 (1.05 – 1.09)	<0.001	1.05 (1.01 – 1.08)	0.005
Total years in wood industry	1.08 (1.05 – 1.10)	<0.001	1.04 (1.00 – 1.07)	0.057
PPE Usage				
Frequency	0.67 (0.51 – 0.89)	0.006	0.74 (0.54 – 1.01)	0.056
Standing Hours	0.91 (0.83 – 0.99)	0.031	0.98 (0.89 – 1.09)	0.732

A fully adjusted binary logistic regression was conducted to evaluate the predictors of high musculoskeletal (MSK) symptom burden. The multivariate model demonstrated exceptional robustness ($\chi^2 = 56.610$, $df = 4$, $p < 0.001$), explaining 19.0% of the variance in MSK pain (Nagelkerke $R^2 = 0.190$), making it the most explanatory model fitted in this study. In contrast to respiratory outcomes, chronic biological factors dominated the MSK risk profile. Age emerged as the primary significant independent predictor: each additional year of life conferred a 5% increase in the odds of reporting severe MSK symptoms (AOR = 1.05; 95% CI: 1.01–1.08; $p = 0.005$). Cumulative occupational loading, measured by total years worked in the wood industry,

demonstrated borderline significance (AOR = 1.04; 95% CI: 1.00–1.07; $p = 0.057$), reflecting the strong collinearity between natural aging and career tenure in this population. Notably, daily standing hours ($p = 0.732$) lost all predictive value after multivariate adjustment. Personal Protective Equipment (PPE) usage exhibited a marginally significant protective trend against MSK pain (AOR = 0.74; $p = 0.056$), though it did not cross the strict threshold for statistical significance."

3.9.3 Predictors of Skin Problems

Table 3.9.3: Binary Logistic Regression — Predictors of Skin Problems

Predictor	Crude OR (95%CI)	p-value	AOR (95% CI)	p-Value
Dust				
Exposure				
Score	0.72 (0.59 – 0.87)	0.001	0.72 (0.57 – 0.91)	0.006
Total years				
in wood				
industry	0.97 (0.95 – 1.00)	0.046	0.97 (0.94 – 1.00)	0.022
Noise				
Exposure				
Score	0.88 (0.79 – 0.98)	0.025	0.98 (0.85 – 1.13)	0.793

A fully adjusted binary logistic regression was conducted to evaluate the independent predictors of high skin symptom burden. The multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 17.933, $df = 3$, $p < 0.001$), explaining 7.3% of the variance in dermatological complaints (Nagelkerke $R^2 = 0.073$). In the adjusted analysis, dust exposure and total years worked in the wood industry were identified as significant independent predictors. The dust exposure score demonstrated a persistent inverse relationship with symptom severity (AOR = 0.72; 95% CI: 0.57–0.91; $p = 0.006$), as did cumulative work duration, where each additional year in the industry was associated with a 3% reduction in the odds of high skin

burden (AOR = 0.97; 95% CI: 0.94–1.00; $p = 0.022$). While ambient noise exposure was significant in the crude analysis (cOR = 0.88; $p = 0.025$), it lost all predictive value after controlling for dust and work duration in the multivariate model ($p = 0.793$).

3.9.4 Predictors of Headache

Table 3.9.4: Binary Logistic Regression — Predictors of Headache

Predictor	Crude OR (95% CI)	p-value	AOR (95% CI)	p-Value
Noise Exposure				
Score	0.87 (0.79 – 0.97)	0.01	0.89 (0.80 – 0.99)	0.026
Age (years)	1.02 (1.00 – 1.04)	0.028	1.02 (1.00 – 1.04)	0.077

A binary logistic regression was conducted to evaluate the independent predictors of high headache burden. The adjusted multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 9.803, $df = 2$, $p = 0.007$), explaining approximately 3.7% of the variance in headache frequency (Nagelkerke $R^2 = 0.037$). In the crude analysis, both age and noise exposure were significantly associated with headache outcomes. However, after multivariate adjustment, noise exposure emerged as the sole significant independent predictor (AOR = 0.89; 95% CI: 0.80–0.99; $p = 0.026$). The inverse adjusted odds ratio reflects the directionality of the specific ordinal scale used to capture noise exposure. While age demonstrated a positive association in the univariate model (cOR = 1.02; $p = 0.028$), it was attenuated to marginal significance once noise exposure was accounted for in the final model (AOR = 1.02; 95% CI: 1.00–1.04; $p = 0.077$).

3.9.5 Educational Attainment and PPE Compliance

Table 3.9.5: Spearman's Correlation — Educational Level and PPE Usage (N = 376)

Variable Pair	rs	p-value	N
Educational level — PPE usage frequency	0.143	0.005	376

A statistically significant positive correlation was identified between workers' educational attainment and their self-reported PPE usage frequency ($r_s = 0.143$, $p = 0.005$), indicating that higher levels of formal education are associated with improved occupational safety compliance. While the effect size is modest, this finding has meaningful programmatic implications: it suggests that targeted educational and health literacy interventions — rather than PPE provision alone — may be essential components of effective compliance strategies in this workforce.

3.10 Quality of Life Impact

The occupational morbidity documented in this study translated directly into functional limitations outside the workplace. When asked if their work-related health problems affected their ability to perform household chores or enjoy their personal life, a striking majority reported a negative impact. Specifically, 61.2% of workers experienced a 'slight' reduction in their quality of life, while 6.9% reported that their occupational injuries 'significantly' impaired their personal and home life. Only 31.9% of the workforce reported that their home life remained completely unaffected by their occupational health status.